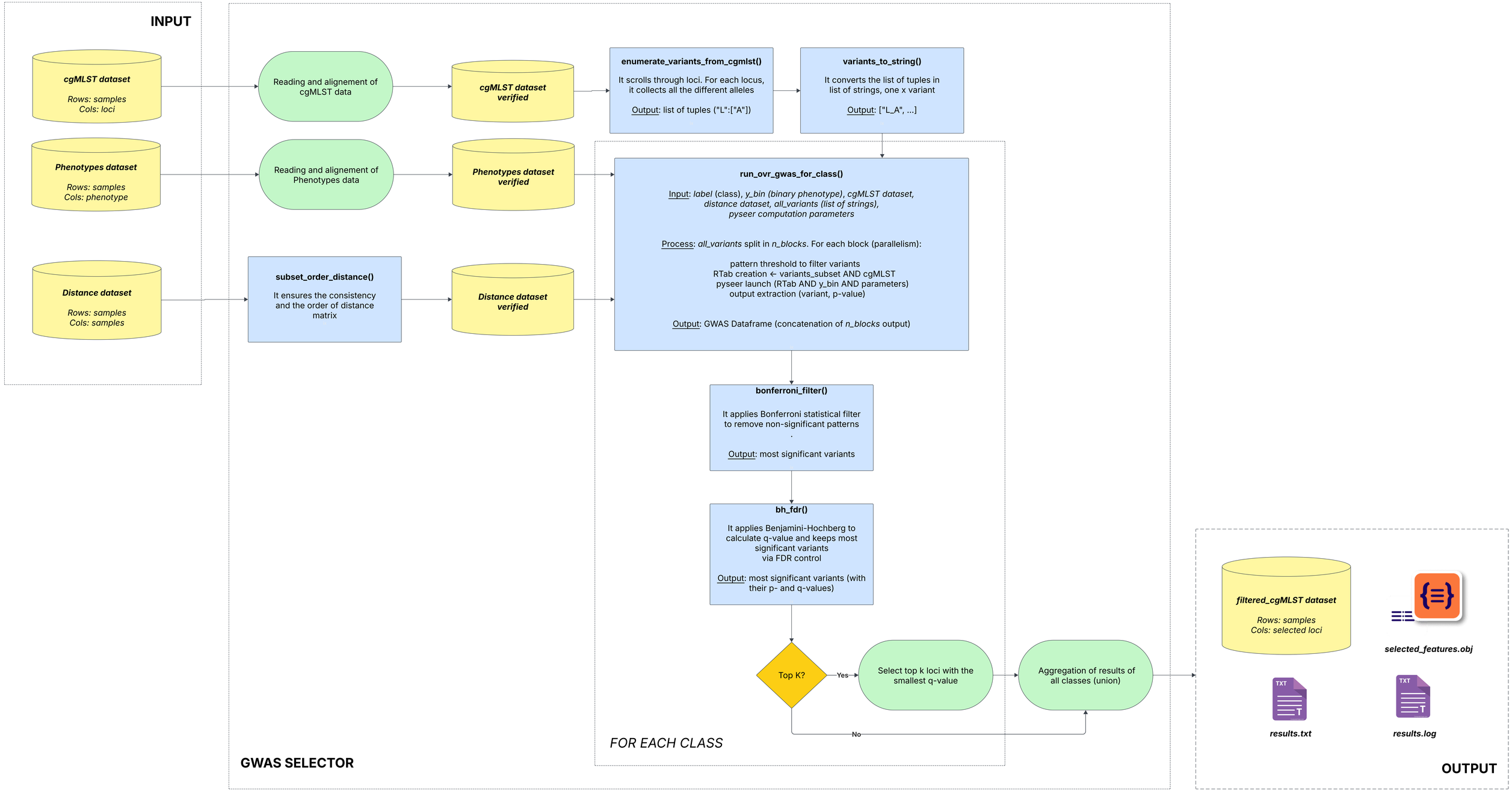
INPUT
cgMLST dataset
Rows: samples
Cols: loci
Phenotypes dataset
Rows: samples
Cols: phenotype
Distance dataset
Rows: samples
Cols: samples
Reading and alignement of cgMLST data
Reading and alignement of Phenotypes data
subset_order_distance()
It ensures the consistency and the order of distance matrix
cgMLST dataset verified
Phenotypes dataset verified
Distance dataset verified
enumerate_variants_from_cgmlst()
It scrolls through loci. For each locus, it collects all the different alleles
Output: list of tuples ("L":["A"])
variants_to_string()
It converts the list of tuples in list of strings, one x variant
Output: ["L_A", ...]
run_ovr_gwas_for_class()
Input: label (class), y_bin (binary phenotype), cgMLST dataset, distance dataset, all_variants (list of strings), pyseer computation parameters
Process: all_variants split in n_blocks. For each block (parallelism):
pattern threshold to filter variants
RTab creation ← variants_subset AND cgMLST
pyseer launch (RTab AND y_bin AND parameters)
output extraction (variant, p-value)
Output: GWAS Dataframe (concatenation of n_blocks output)
bonferroni_filter()
It applies Bonferroni statistical filter to remove non-significant patterns
.
Output: most significant variants
bh_fdr()
It applies Benjamini-Hochberg to calculate q-value and keeps most significant variants via FDR control
Output: most significant variants (with their p- and q-values)
Top K?
Yes
No
Select top k loci with the smallest q-value
Aggregation of results of all classes (union)
FOR EACH CLASS
GWAS SELECTOR
filtered_cgMLST dataset
Rows: samples
Cols: selected loci
selected_features.obj
TXT
results.txt
TXT
results.log
OUTPUT